Introduction to Computer Science and Programming

Introduction to EduBlocks

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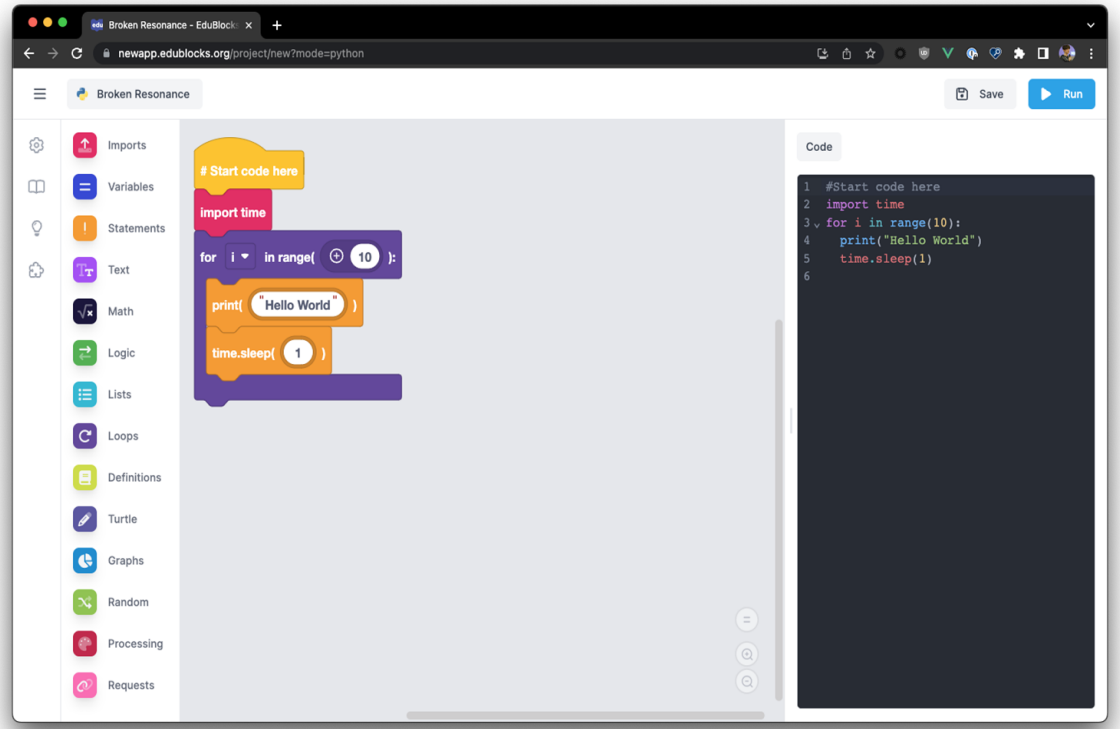
What is EduBlocks?

EduBlocks is a **free application** available online at **edublocks.org**

It was started by a **12 year-old student** who wanted to help others learn how to code

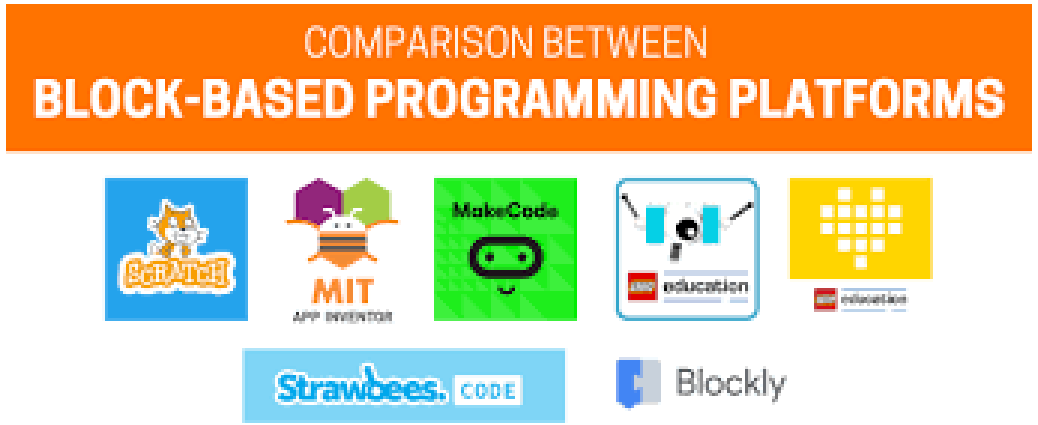
It **runs in the browser** and works on all operating systems

Blocks are transformed in pure **Python** and then executed



EduBlocks: More Than Just a Game!

- At first glance, **EduBlocks** may look like a game — colorful blocks, drag-and-drop coding, and a playful interface. But in reality, it's built on the concept of **block-based programming**, a paradigm also used beyond education
- Tools such as **Blockly**, **Node-RED**, and **App Inventor** adopt this approach in areas like **IoT, automation, web development, and data visualization**, proving that visual programming can be powerful and practical
- We will use **EduBlocks** as an introduction to programming — an accessible way to learn core coding concepts before moving to text-based languages like **C++**



Getting Started

To write code, we'll need to create a project

Open a browser

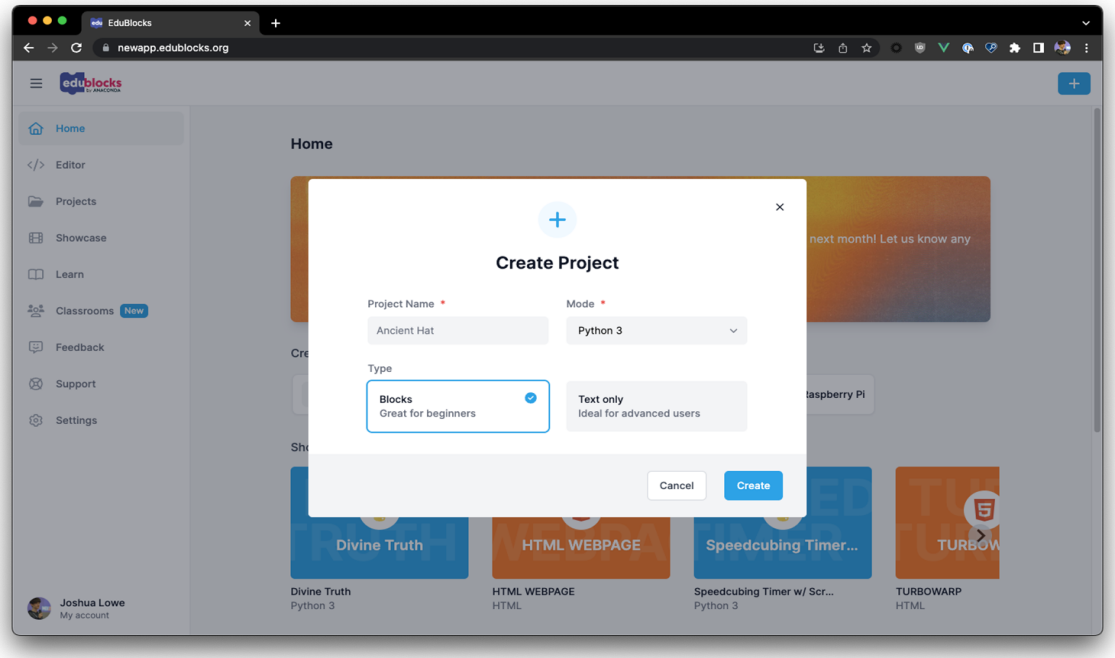
Go to app.edublocks.org/editor

Select a mode

We'll be using Python, so select "Python 3"

Get started!

Rename your project to "FirstProject" and click **Create**!



The Interface

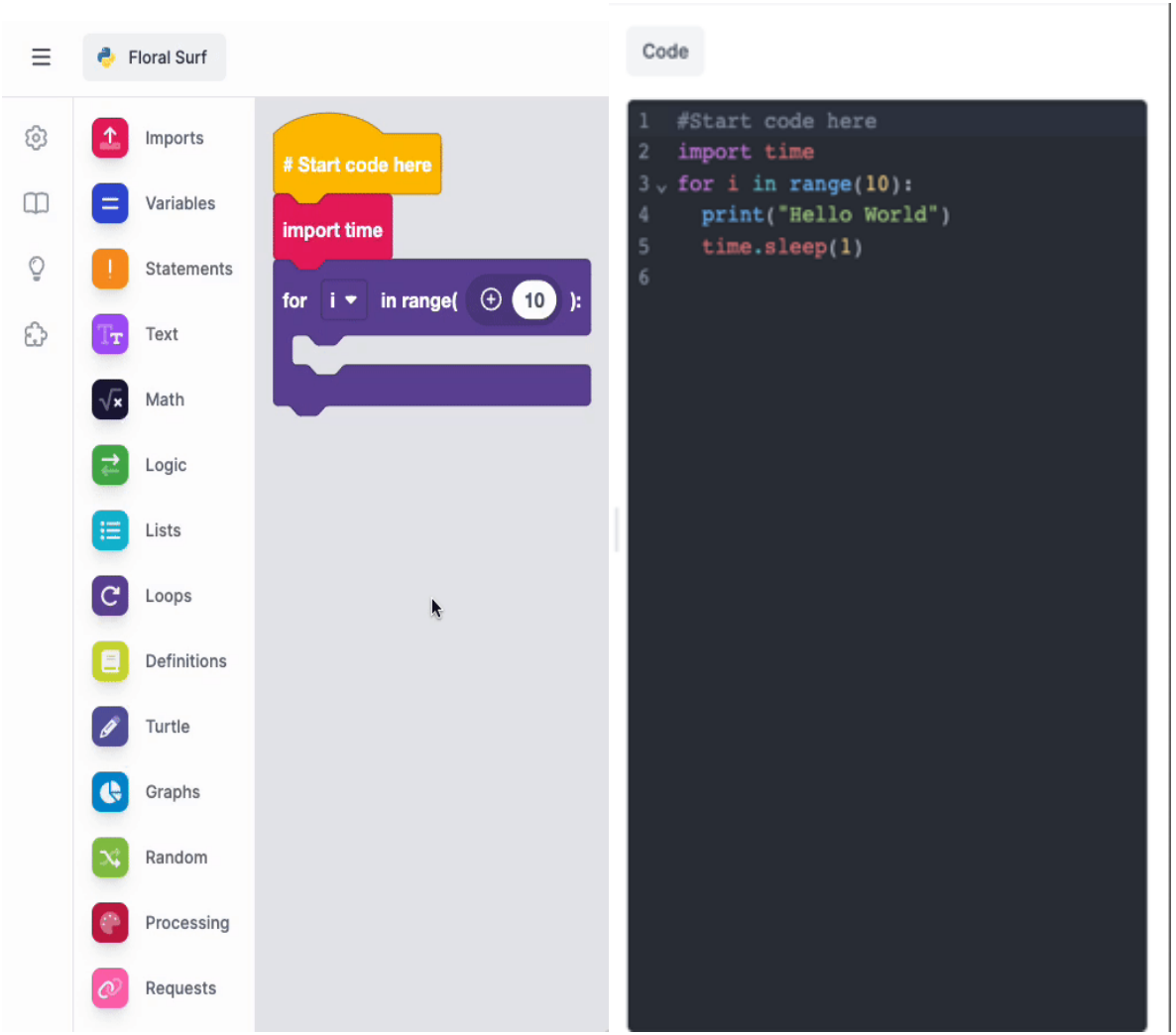
The EduBlocks Interface is simple

On the left we have all of the blocks we can use to write code

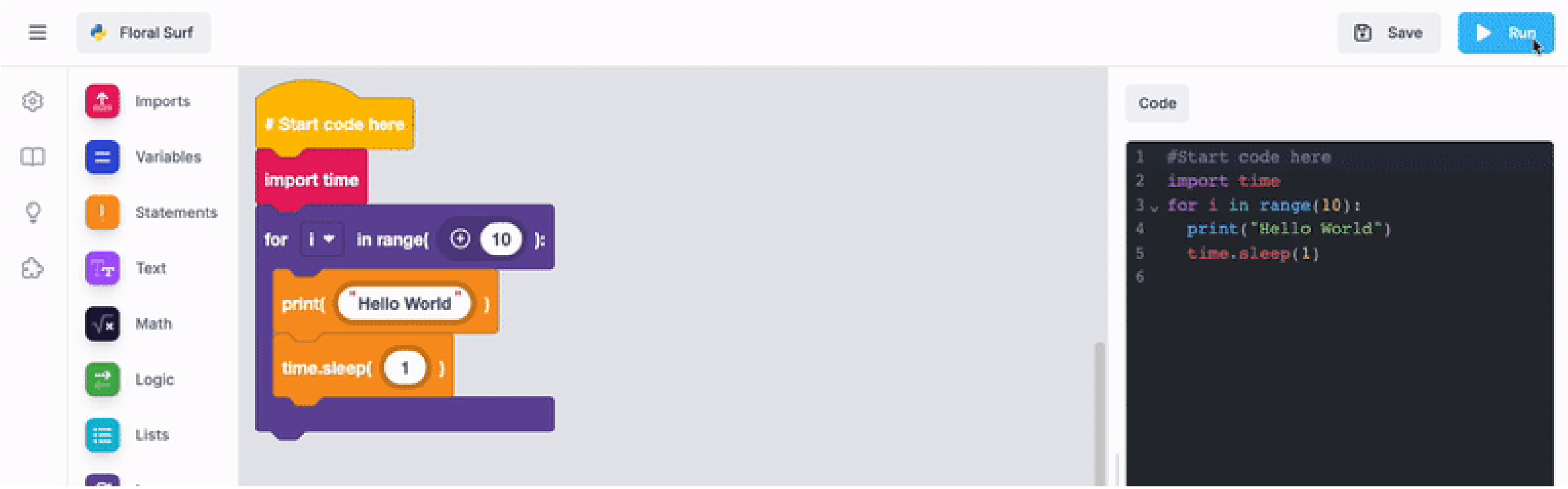
The blocks are places in the coding area on the left of the screen

Blocks are organised in the menu **by their function**

On the right we can see the generated Python code



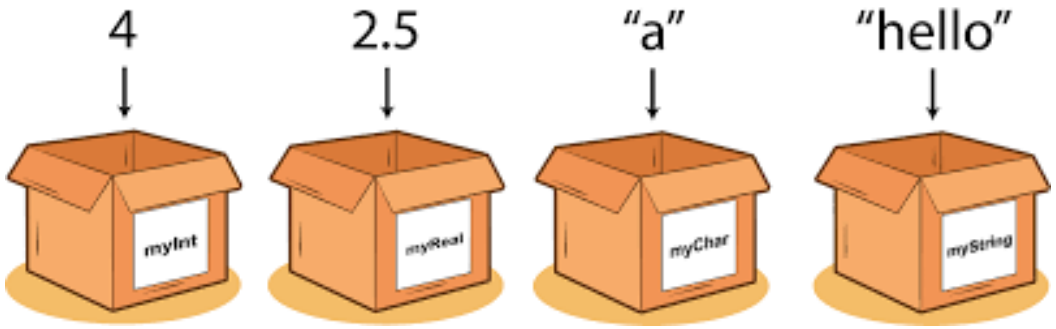
Run the code



To run our code we click the Blue run button. This shows the output on the right of the screen

Variables

- Variables are **widely used during the description of an algorithm!**
- Variables are names (possibly meaningful) used to refer to some memory locations of a computer
- Variables are **containers that can hold values**
- **The value of a variable is not fixed but can change during the execution of a program!**
- **A Variable has a type**
  - **E.g., int myInt**



Basic data types in EduBlock

At the basic level there are three data types in EduBlock

String

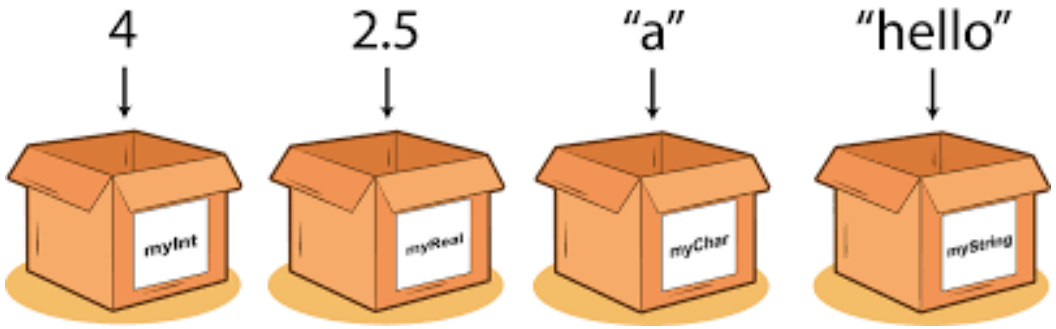
A sequence of characters, including punctuation, numbers and letters.

Integer

A number with no decimal place

Float

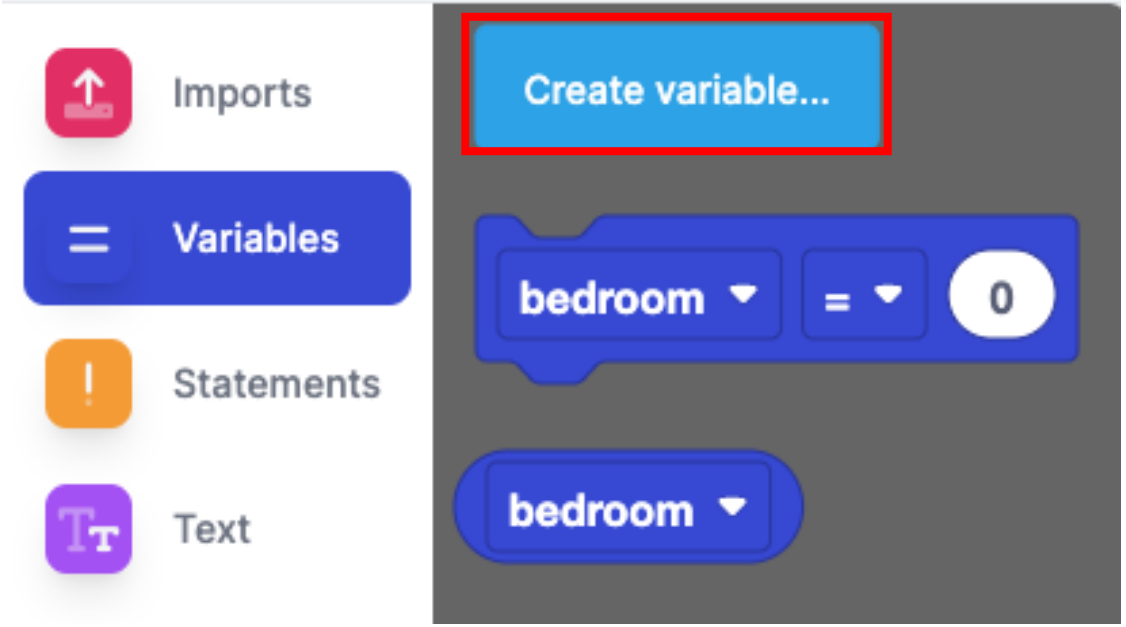
A number with a decimal place



Variable Creation

We first create a variable:

- Go to variables
- Click on “Create Variable”
- Call the variable “bedroom” (for example)
- New blocks will appear
- We can drag and drop them in the code area



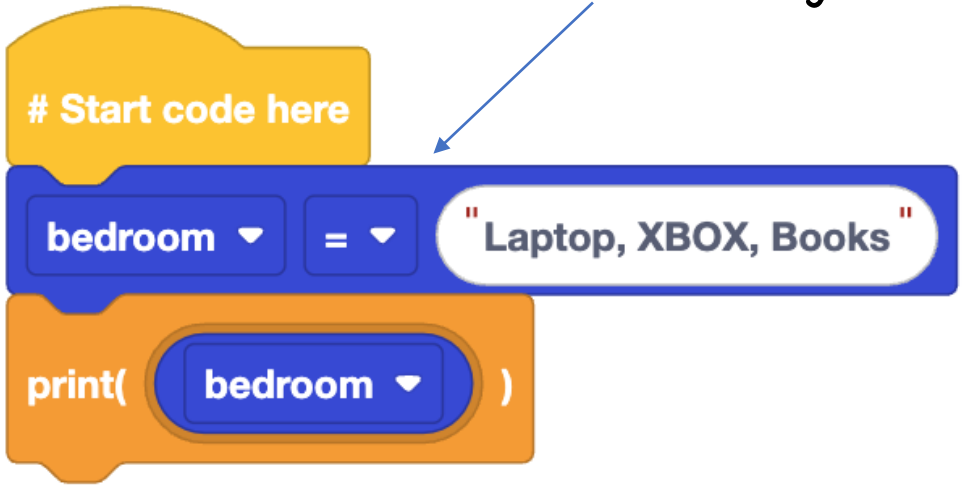
Assignment and Print

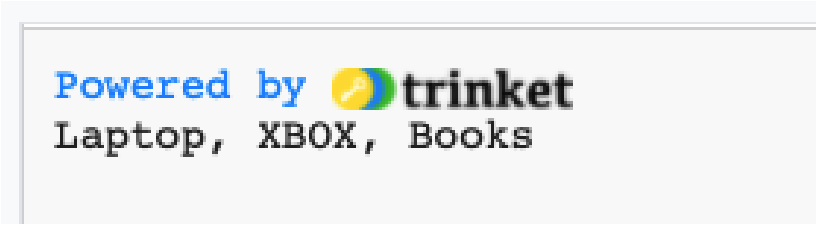
Here we fill the bedroom variable with items in our bedroom

Then we call the variable by name and print the contents

Variables can store any data type

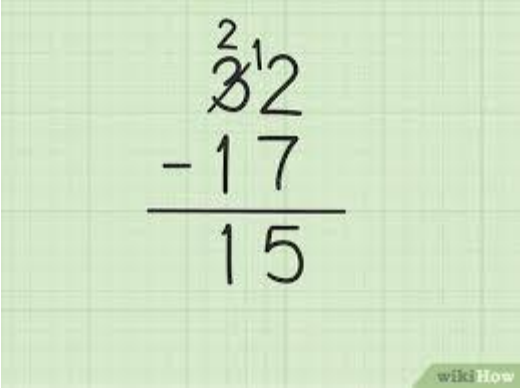
Variable assignment





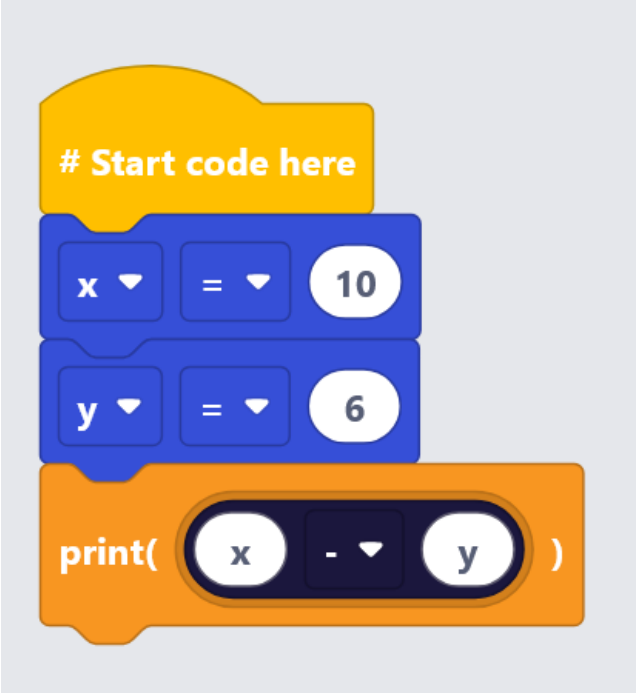
Print the value of the variable
(it corresponds to Display in our Pseudocode)

Using variables



Exercise: Create a simple **EduBlock program** with two variables x (equal to 10) and y (equal to 6) and print x – y (the result will be 4)

- Set the value of x equal to 10
- Set the value of y equal to 6
- Print x minus y



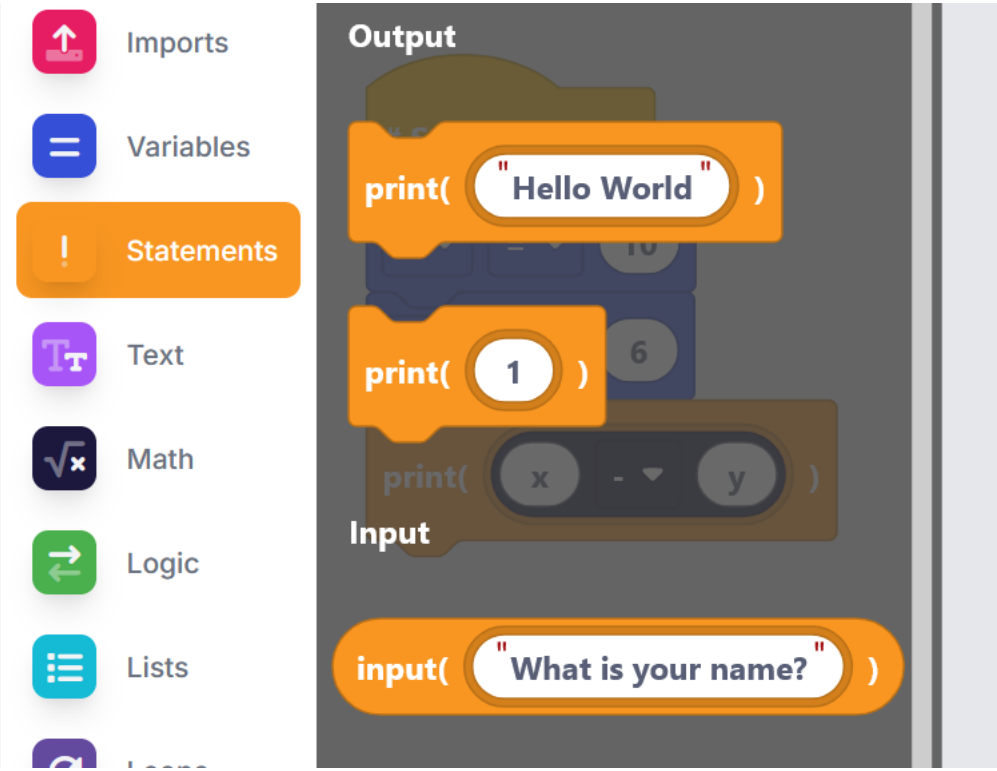
How can we use the input?

User input is a way that the computer can ask a question

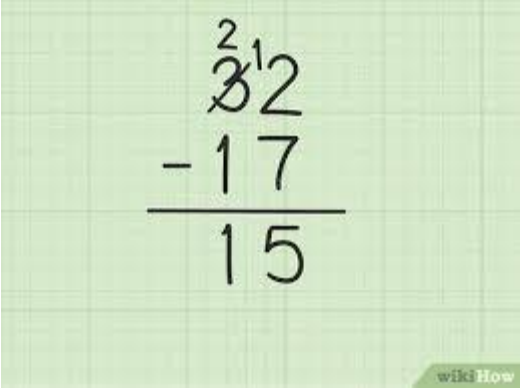
Insert a value?

Traditionally user input refers to keyboard input

The input block is found in the **statements section**

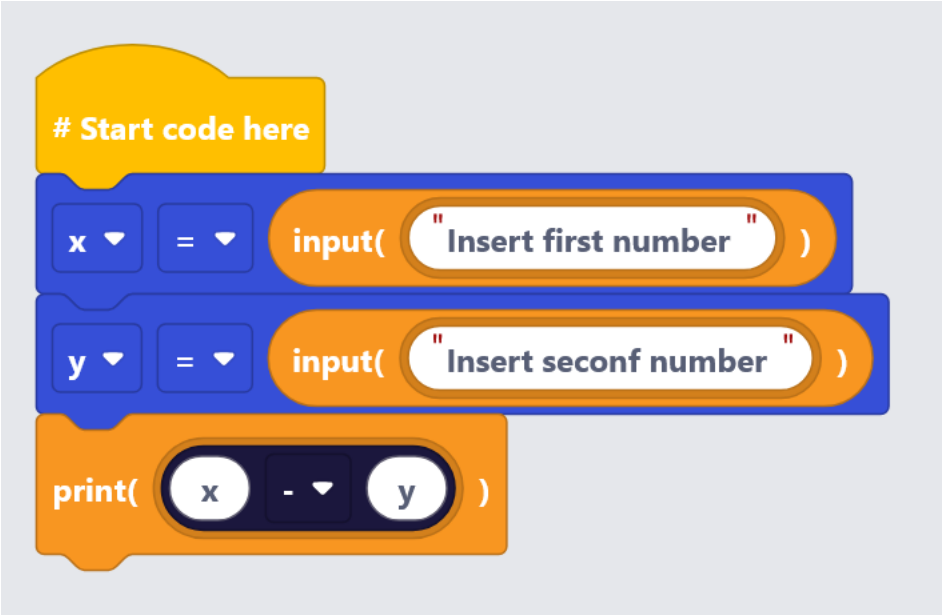


Using variables and Input



Exercise: Create a simple **EduBlock program** with two variable x and y but now the **values are read from the user input** (entered from the keyboard)

- Get x
- Get y
- Print x minus y



If we run the code and enter 10 and 6, what is the output?

```
Insert first number 10
Insert seconf number 6
TypeError: unsupported operand type(s) for -: 'str' and 'str'
```

What happened?

There was an error!

But what does this mean?

```
Insert first number 10
Insert seconf number 6
TypeError: unsupported operand type(s) for -: 'str' and 'str'
```

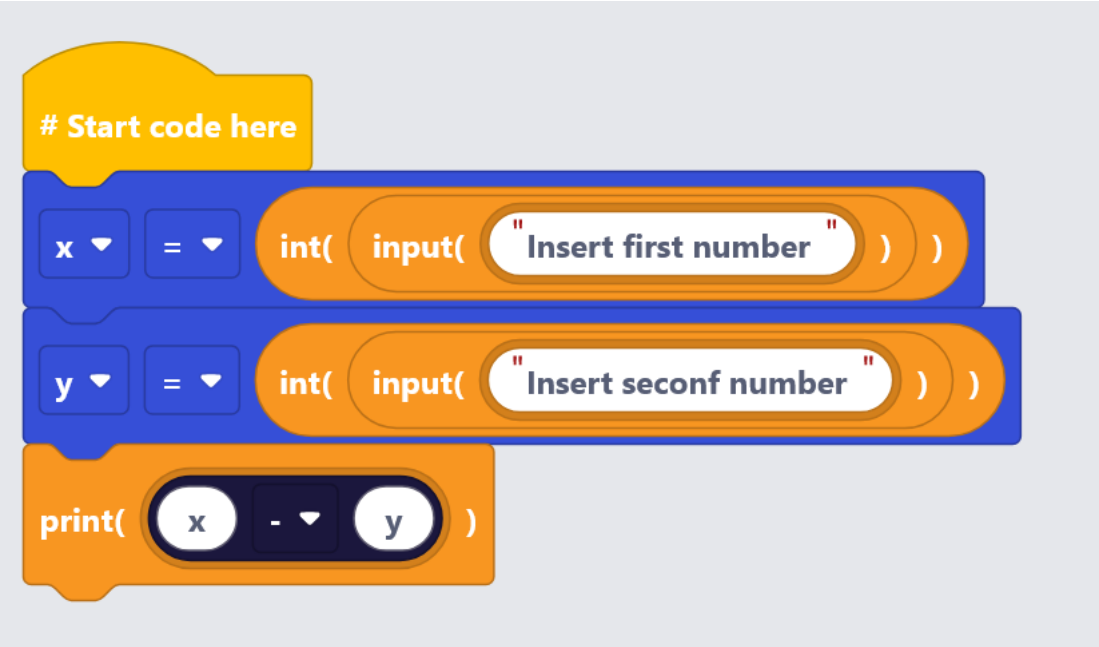
- Data types tell a computer how we want to use the data

- A **user input captures the number as a string**, a sequence of characters
- We are trying to subtract two strings, an operation that is not allowed!!!

How can we fix the error?

We need to convert the strings to integers

- Use the int(1) block from statements
- Wrap it around the input() block



Now it works!!!

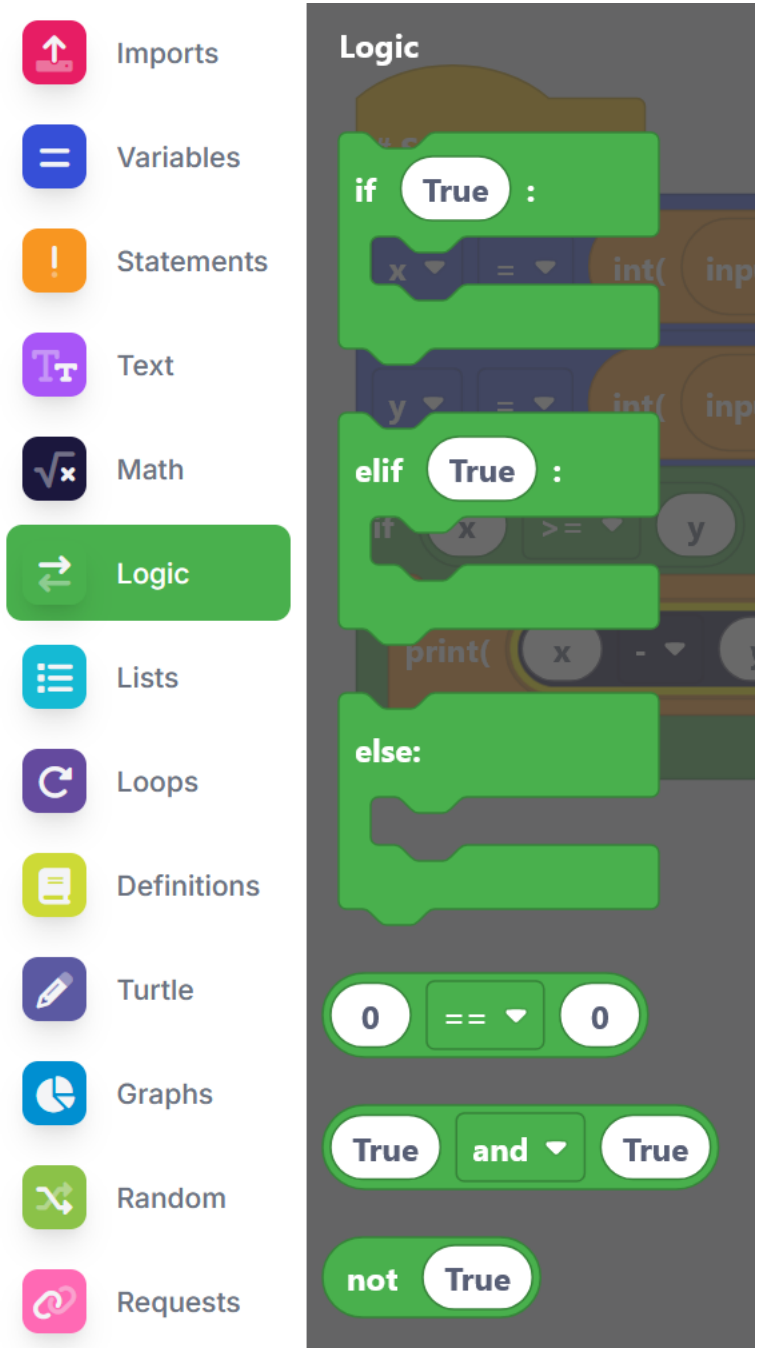
```
Insert first number 10
Insert seconf number 6
4
```

If and if-else

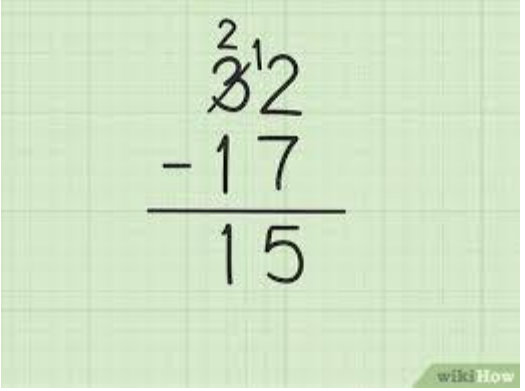
How do we make a decision?

We have to use if and if-else blocks

If and if-else blocks are in the Logic session

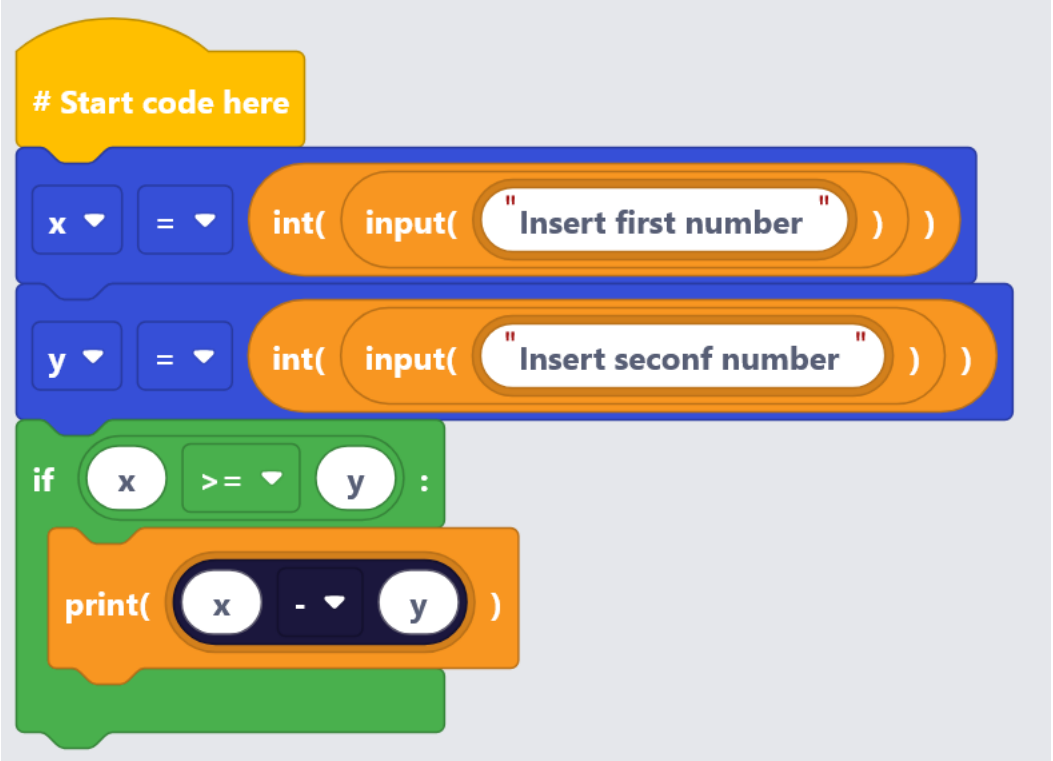


Using variables, Input and If



Exercise: Create a simple **EduBlock program** with two variables, x and y, where the values are read from user input (typed from the keyboard). The operation should be performed only if x is greater than or equal to y; otherwise, it should not be executed

- Get x
- Get y
- If x >= y
    - Print x minus y



Logical Thinking

Condition Block

We can have different kind of conditions

==

Values must be equal

!=

Values must NOT be equal

<

Left value is less than right value

<=

Left value is less than or equal to right value

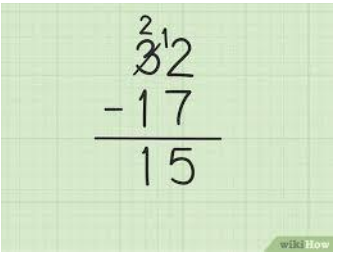
>

Left value is greater than right value

>=

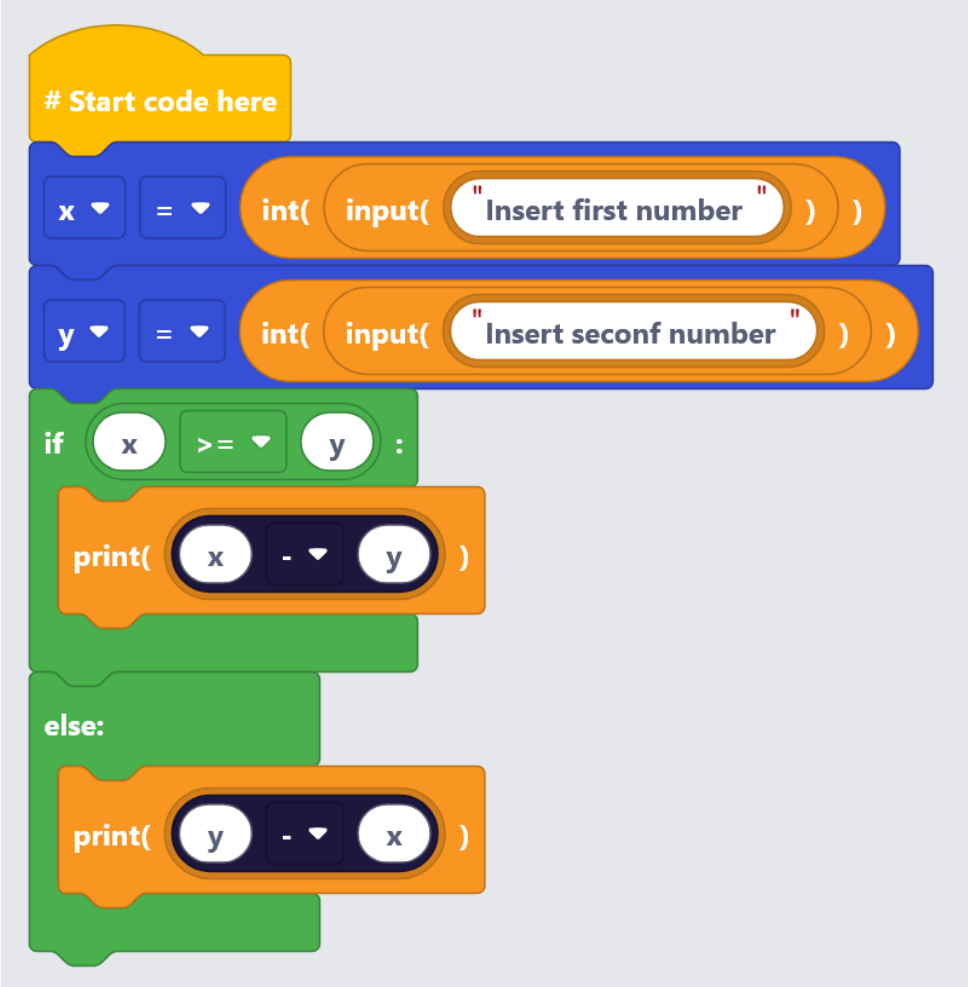
Left value is greater than or equal to right value

Using variables, Input and If-else



Exercise: Create a simple **EduBlock program** with two variables, x and y, where the values are read from user input (typed from the keyboard). The program should always subtract the smaller number from the larger one

- Get x
- Get y
- If x >= y
  - Print x minus y
- Else
  - Print y minus x



If-else if and else

Here if we answer **yes** to the first question then the computer will say “Hi, Ed”

If we answer **no**, it will ask us another question

But if we answer no both questions, it will just say “I don’t know you, but hello!”

```
if input( "Is your name Ed?" ) == "yes" :
    print( "Hi, Ed!" )
elif input( "Hmm, maybe Jasmine?" ) == "yes" :
    print( "Hi, Jasmine!" )
else:
    print( "I don't know you, but hello!" )
```

Exercise 1: Final Grade

- First, **write the pseudocode** for a program that:
  - asks the user to enter **four integer grades** (each between 0 and 30),
  - calculates the **average** of the four grades,
  - and prints **Passing** or **Failing** with a message:
    - **"Passing"** if the grade is **18 or higher**,
    - **"Failing"** if it is **below 18**.
- Then, use **EduBlocks** to build the program following your pseudocode

- **Input/Output:**
  - **18, 20, 18, 20 -> Passing**
  - **10, 8, 12, 18 -> Failing**

Pseudocode (1) : Final Grade

Version 1:

1. *“Input a set of 4 marks”*
2. *“Calculate their average by summing and dividing by 4”*
3. *if average is below 18 then Print ‘FAILING’*

   else Print ‘PASSING’

Pseudocode (2) : Final Grade

Version 1:

1. *"Input a set of 4 marks"*
2. *"Calculate their average by summing and dividing by 4"*
3. *if average is below 18 then Print 'FAIL' else Print 'PASS'*

We introduce variables and use if-then

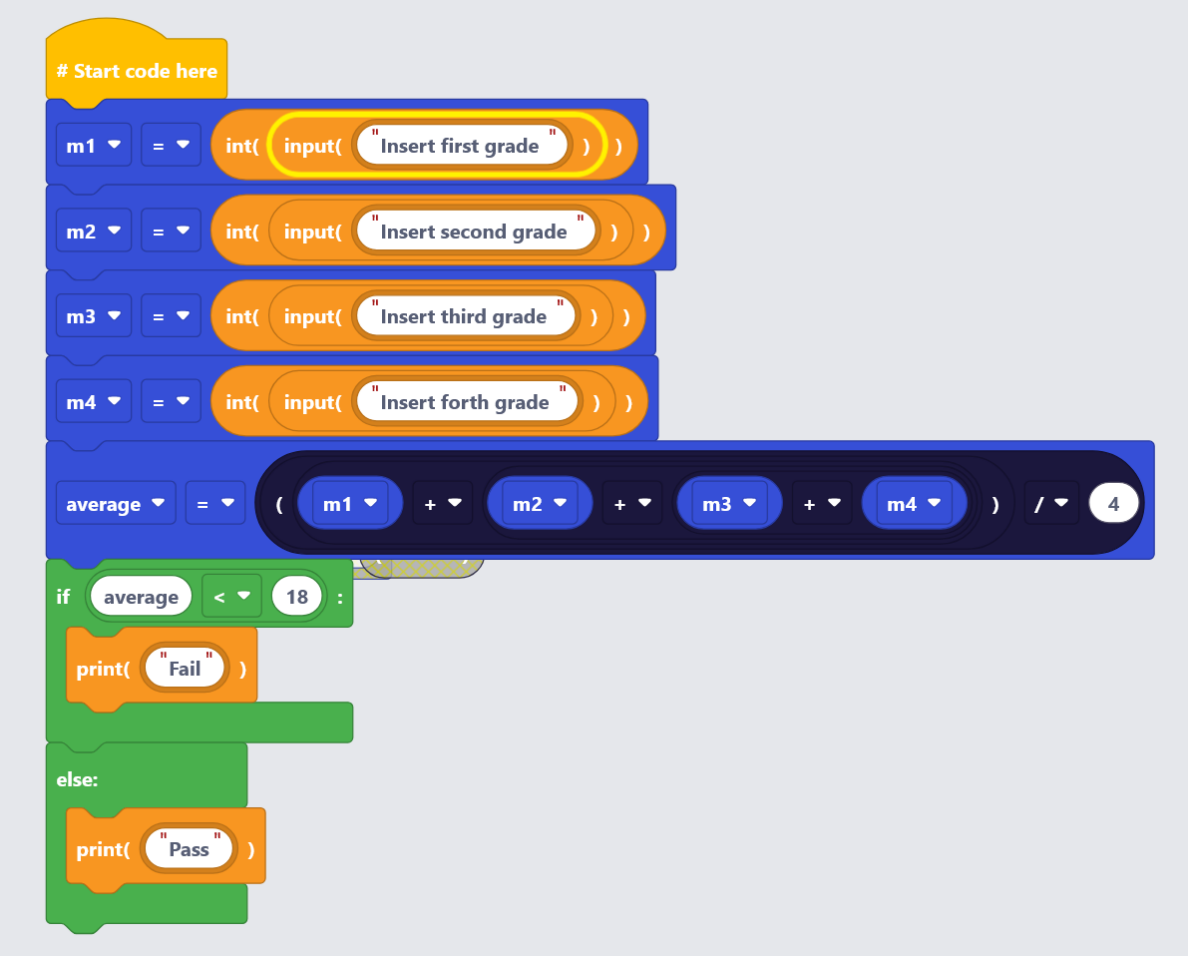
Version 2 (refinement):

1. **GET** m1, m2, m3, m4
2. average ← (m1+m2+m3+m4)/4
3. **IF** (average < 18) **THEN**
4. **DISPLAY** "FAILING"
5. **ELSE**
6. **DISPLAY** "PASSING"
7. **ENDIF**

EduBlock Implementation: Final Grade

Version 2 (refinement):

1. **GET** m1, m2, m3, m4
2. average ← (m1+m2+m3+m4)/4
3. **IF** (average < 18) **THEN**
4. **DISPLAY** “FAILING”
5. **ELSE**
6. **DISPLAY** “PASSING”
7. **ENDIF**



Exercise 2: Simple Calculator

- Before coding, **write the pseudocode** for a simple calculator that:
  - asks the user to choose an operator (+, -, *, /),
  - asks for two numbers,
  - performs the correct calculation,
  - and shows the result or “Unknown operator” if the input is invalid (e.g, 3 % 2).
- Then, use **EduBlocks** to build the program following your pseudocode

Input/Output:

- ‘-’, 10, 3 -> 7
- ‘%’, 7, 2 -> Unknown operator

Pseudocode: Simple Calculator

Version 1:

“Input two numbers and an operator (+, –, *, /)”
“Depending on the operator, perform the correct calculation”
“If the operator is not valid, print ‘Unknown operator’”
“Display the result of the operation”

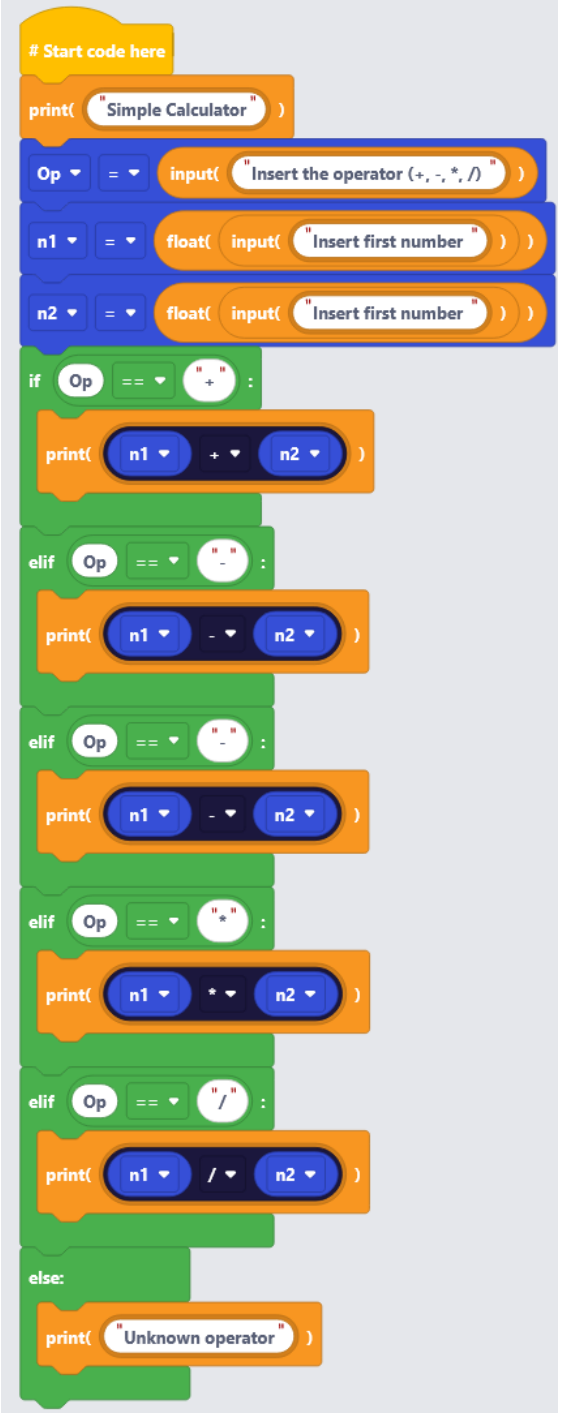
Version 2 (refinement):

```
GET operator
GET n1
GET n2
IF operator = "+" THEN
  DISPLAY n1 + n2
ELSE IF operator = "-" THEN
  DISPLAY n1 - n2
ELSE IF operator = "*" THEN
  DISPLAY n1 * n2
ELSE IF operator = "/" THEN
  DISPLAY n1 / n2
ELSE
  DISPLAY "Unknown operator"
ENDIF
```

EduBlock Implementation: Simple Calculator

Version 2 (refinement):

1. GET operator
2. GET n1
3. GET n2
4. IF operator = "+" THEN
5. DISPLAY n1 + n2
6. ELSE IF operator = "-" THEN
7. DISPLAY n1 - n2
8. ELSE IF operator = "*" THEN
9. DISPLAY n1 * n2
10. ELSE IF operator = "/" THEN
11. DISPLAY n1 / n2
12. ELSE
13. DISPLAY "Unknown operator"
14. ENDIF

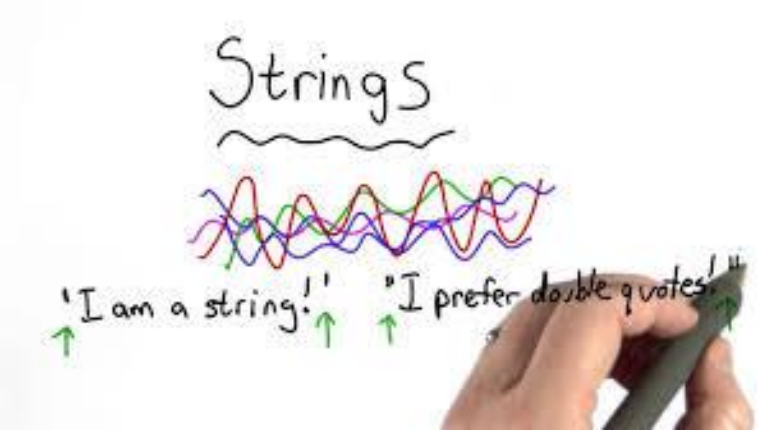


Exercise 3: Count Vowels

- Write the **pseudocode**, then create an **EduBlocks** program that:
  - asks the user to enter a word or sentence (i.e., a string)
  - counts how many vowels it contains,
  - and displays the total number of vowels.

Input/Output:

- “house” -> 3
- “car” -> 1
- “The cat is on the table” -> 7



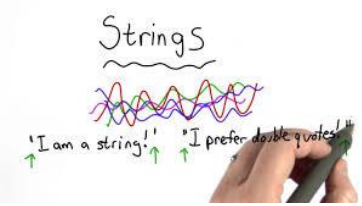
Pseudocode: Count Vowels

Version 1:

“Ask the user to input a word or a sentence”
“Convert the text to lowercase”
“Count how many times each vowel appears”
“Add all the counts together”
“Display the total number of vowels”

Version 2 (refinement):

```
GET text
text = LOWERCASE(text)
a = COUNT(text, "a")
e = COUNT(text, "e")
i = COUNT(text, "i")
o = COUNT(text, "o")
u = COUNT(text, "u")
total = a + e + i + o + u
DISPLAY total
```



EduBlock Implementation: Count Vowels

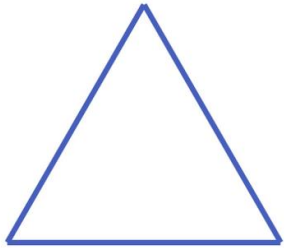
Version 2 (refinement):

```
GET text
text = LOWERCASE(text)
a = COUNT(text, "a")
e = COUNT(text, "e")
i = COUNT(text, "i")
o = COUNT(text, "o")
u = COUNT(text, "u")
total = a + e + i + o + u
DISPLAY total
```

Exercises

1. Calculate the Area of a Rectangle:

Problem: Write the **Pseudocode** and **EduBlocks** program that calculates the area of a rectangle given its length and width
Input: Length = 10, Width = 5, **Output**: Area = 50



2. Grade Calculator:

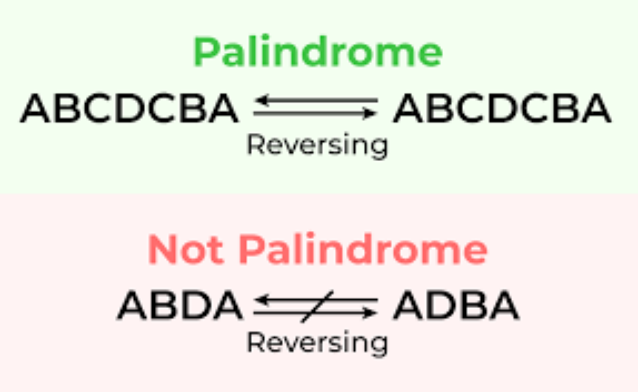
Problem: Write the **Pseudocode** and **EduBlocks** program that takes a percentage as input and outputs the corresponding letter grade (A, B, C, D, F).
Input: 85%, **Output**: B

Grading Scale (example):

- A: 90% - 100%
- B: 80% - 89%
- C: 70% - 79%
- D: 60% - 69%
- F: Below 60%

3. Palindrome Checker

Problem: Write the **Pseudocode** and **EduBlocks** program that:

- asks the user to enter a word or phrase,
- removes spaces and converts the text to lowercase,
- checks if the text reads the same backward and forward,
- and displays whether it **is** or **is not** a palindrome.

Input: “Kayak“, **Output**: Palindrome